B - Spiral Galaxy

分值：200200200 分

Problem Statement

There is a grid with HHH rows and WWW columns. The cell at the iii-th row from the top and jjj-th column from the left is denoted as cell $(i,j)(i,j)(i,j)$.

有一个 HHH 行 WWW 列的网格。从上往下第 iii 行、从左往右第 jjj 列的单元格记为 $(i,j)(i,j)(i,j)$。

Each cell of the grid is colored white or black. The information of the grid is given by HHH strings $S1,S2,\dots,SHS_1,S_2,\dots,S_HS1,S2,\dots,SH$ each of length WWW: cell $(i,j)(i,j)(i,j)$ is white if the jjj-th character of SiS_iSi is `.`, and black if it is `#`.

网格的每个单元格被染为白色或黑色。网格的信息由 HHH 个长度为 WWW 的字符串 $S1,S2,\dots,SHS_1,S_2,\dots,S_HS1,S2,\dots,SH$ 给出：若 SiS_iSi 的第 jjj 个字符为 `.`，则单元格 $(i,j)(i,j)(i,j)$ 为白色，若为 `#` 则为黑色。

Find the number of rectangular regions of the grid that are point-symmetrically colored.

求网格中满足染色是中心对称的矩形区域的数量。

More formally, find the number of integer tuples $(h1,h2,w1,w2)(h_1,h_2,w_1,w_2)(h1,h2,w1,w2)$ satisfying all of the following conditions:

更正式地，求满足以下所有条件的整数元组 $(h1,h2,w1,w2)(h_1,h_2,w_1,w_2)(h1,h2,w1,w2)$ 的数量：

- $1 \le h1 \le h2 \le H1 \le h_1 \le h_2 \le H1 \le h1 \le h2 \le H$
- $1 \le w1 \le w2 \le W1 \le w_1 \le w_2 \le W1 \le w1 \le w2 \le W$
- For all integers i,ji,ji,j satisfying $h1 \le i \le h2h_1 \le i \le h_2h1 \le i \le h2$ and $w1 \le j \le w2w_1 \le j \le w_2w1 \le j \le w2$, cell $(i,j)(i,j)(i,j)$ and cell $(h1+h2-i,w1+w2-j)(h_1+h_2-i,w_1+w_2-j)(h1+h2-i,w1+w2-j)$ have the same color.
- $1 \le h1 \le h2 \le H1 \le h_1 \le h_2 \le H1 \le h1 \le h2 \le H$
- $1 \le w1 \le w2 \le W1 \le w_1 \le w_2 \le W1 \le w1 \le w2 \le W$
- 对所有满足 $h1 \le i \le h2h_1 \le i \le h_2h1 \le i \le h2$ 和 $w1 \le j \le w2w_1 \le j \le w_2w1 \le j \le w2$ 的整数 i,ji,ji,j，单元格 $(i,j)(i,j)(i,j)$ 和单元格 $(h1+h2-i,w1+w2-j)(h_1+h_2-i,w_1+w_2-j)(h1+h2-i,w1+w2-j)$ 颜色相同。

Constraints

- $1 \le H,W \le 101 \le H,W \le 101 \le H,W \le 10$
- HHH and WWW are integers.

- SiS_iSi is a string of length WWW consisting of `.` and `#`.
- $1 \le H, W \le 101 \le H, W \le 101 \le H, W \le 10$
- HHH 和 WWW 是整数。
- SiS_iSi 是长度为 WWW 的字符串，由 `.` 和 `#` 组成。

Input

The input is given from Standard Input in the following format:

输入按以下格式从标准输入给出：

HHH WWW
$S1S_1S1$
$S2S_2S2$
$\vdots\vdots\vdots$
SHS_HSH

Output

Output the answer.

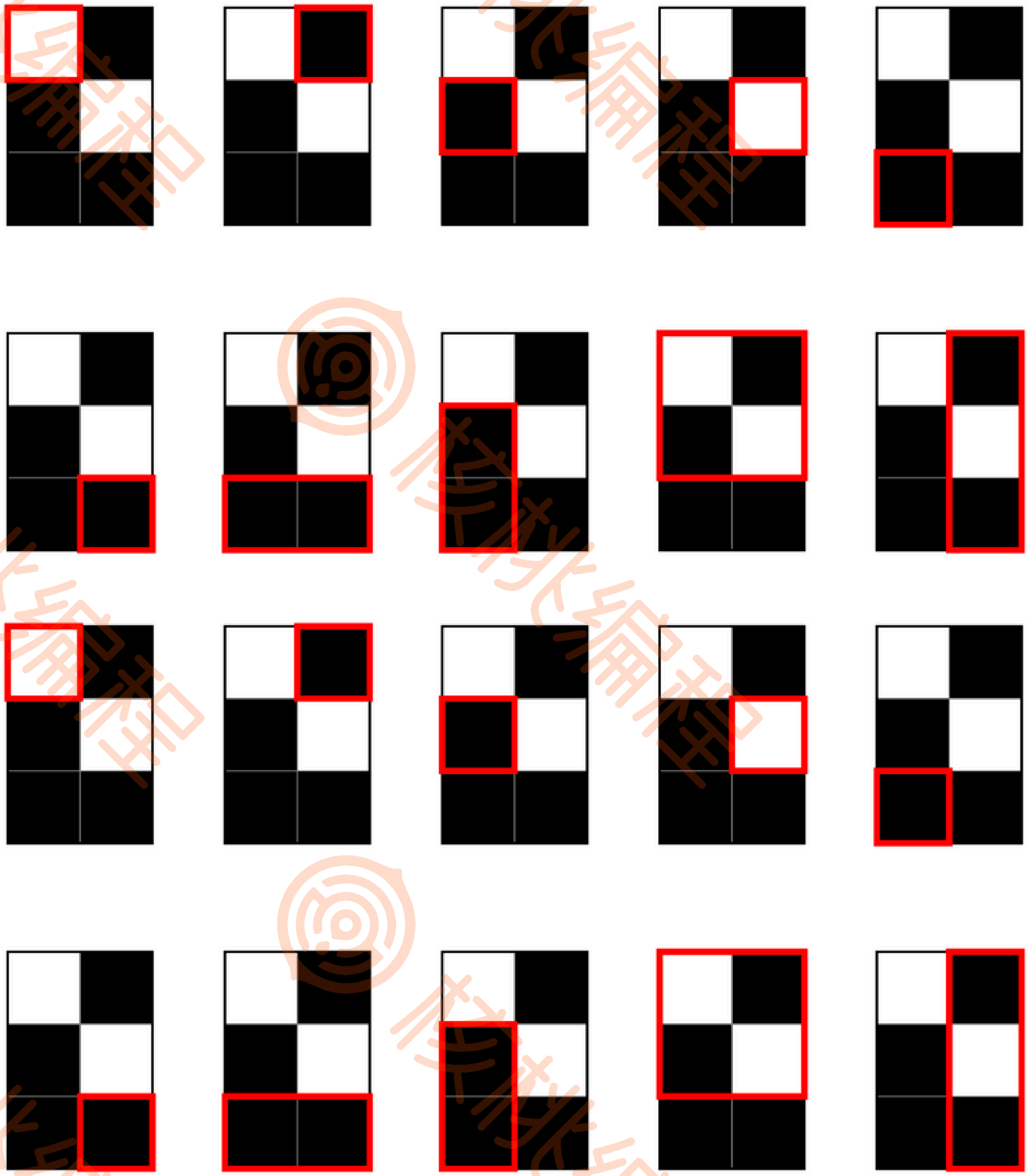
输出答案。

Sample Input 1

```
3 2
.#
#.
##
```

Sample Output 1

```
10
```



As shown in the figure above, the answer is 101010.

如上图所示，答案为 101010。

Sample Input 2

```
4 5
.#.#.
####.
```

```
##..#
....#
```

Sample Output 2

```
54
```